AP CALCULUS AB

Unit 1 • Worksheet 1-3A • Answer key

Worksheet 1-3A • Limits to Infinity and Horizontal Asymptotes

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{x\to\infty} \dfrac{4x^3 - x}{2x^3 + 7x^2}$ is

(A) 0

(B) $\frac{4}{7}$

(C) ∞

(D) 2

Solution

Answer: (D).

Question 2

NO CALCULATOR

$\lim_{x\to-\infty} \dfrac{2x}{\sqrt{x^2 + 5}}$ is

(A) 2

(B) -2

(C) 0

(D) $-\infty$

Solution

Answer: (B). $\sqrt{x^2} = -x$ for $x < 0$.

Question 3

NO CALCULATOR

$\lim_{x\to\infty} \dfrac{3e^x + x^2}{e^x - 1}$ is

(A) 0

(B) 1

(C) 3

(D) ∞

Solution

Answer: (C).

Question 4

NO CALCULATOR

Which line or lines are vertical asymptotes of $h(x) = \dfrac{x+1}{x^2-1}$?

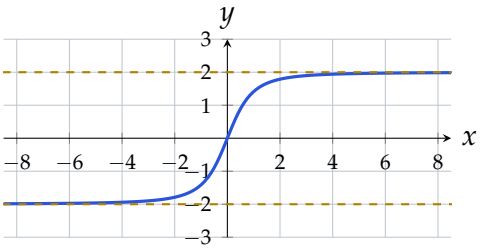
(A) $x = 1$ only
(B) $x = -1$ only
(C) $x = 1$ and $x = -1$
(D) $y = 0$ only

Solution
Answer: (A). Hole at $x = -1$.

Question 5

NO CALCULATOR

The graph of $f(x) = \frac{2x}{\sqrt{x^2+1}}$ is shown. Which are its horizontal asymptotes?



(A) $y = 2$ only
(B) $y = 0$
(C) $y = 2$ and $y = -2$
(D) $x = 2$ and $x = -2$

Solution
Answer: (C).

Question 6 (Free response)

NO CALCULATOR

Let $f(x) = \dfrac{x+2}{x^2-4x}$.

(a) Find the vertical asymptotes and both one-sided limits at each.
(b) Find $\lim_{x\to\pm\infty} f(x)$ and the horizontal asymptote.

Solution
(a) $x = 0$: $\lim_{x\to 0^-} = +\infty$, $\lim_{x\to 0^+} = -\infty$. $x = 4$: $\lim_{x\to 4^-} = -\infty$, $\lim_{x\to 4^+} = +\infty$.
(b) 0; $y = 0$.

Question 7 (Free response) NO CALCULATOR

Find: (a) $\lim_{x\to\infty} \frac{x^3}{e^x}$ (b) $\lim_{x\to\infty} \frac{\ln x}{\sqrt{x}}$ (c) $\lim_{x\to 0^+} \ln x$ (d) $\lim_{x\to \frac{\pi}{2}^-} \tan x$

Solution

(a) 0. (b) 0. (c) $-\infty$. (d) $+\infty$.